Hyperspectral Anomaly Detection Based on Tensor Ring Decomposition With Factors TV Regularization

Maoyuan Feng, Wendong Chen, Yunxiu Yang, Qin Shu, *Member, IEEE*, Hongxin Li, and Yanqin Huang

***Abstract*— Anomaly detection in the hyperspectral image (HSI) has gradually become a hot topic in remote sensing. Recently, some tensor-based methods have been proposed to improve detection performance by exploiting the characteristic of HSI data existing in the inherent multidimensional. However, the existing tensor-based methods may only partially use the prior properties in both spatial and spectral dimensions. In this article, we proposed a novel tensor ring (TR) decomposition with factors total variation (TV) regularization model for hyperspectral anomaly detection (TRDFTVAD). First, raw HSI data are decomposed into background and anomaly tensors. The TR decomposition is adopted to exploit the low-rank property of the background existing in both spatial and spectral dimensions. Then, the TV regularization is designed on the three dimensions of the background tensor to explore the piecewise smoothness of the background existing in both spatial and spectral dimensions. Further, this TV regularization is transferred to each factor tensor by exploiting the relationship between the background tensor and each factor tensor. Next, the $l_{2,1}$ norm regularization is designed on the anomaly tensor to exploit the group sparsity of anomaly pixels. Finally, the alternating direction method of multipliers (ADMM) scheme is adopted to update the involved variables. Experimental results validated on several real hyperspectral datasets demonstrate the effectiveness of the proposed algorithm.**

***Index Terms*— Hyperspectral anomaly detection, low-rank tensor, tensor ring (TR) decomposition, total variation (TV) regularization.**

NOMENCLATURE

x	Scalar.
$\mathbf{x}$	Vector.
$\mathbf{X}$	Matrix.
$\mathcal{X}$	Tensor.
$\mathbf{X}_{ij}$	(i, j)th element.
$\mathcal{X}(i, j, :)$	$(i.j)$th tube.
$\lvert x\rvert$	Absolute value.
$(\cdot)^T$	Transpose operator.
$\lVert\mathbf{x}\rVert_2$	$\sqrt{\sum_i x_i^2}$.
$\lVert\mathcal{X}\rVert_1$	$\sum_{ijk}\lvert\mathcal{X}_{ijk}\rvert$.
$\lVert\mathcal{X}\rVert_{2,1}$	$\sum_{ij}\lVert\mathcal{X}(i, j, :)\rVert_2$.
$\lVert\mathcal{X}\rVert_F$	$\sqrt{\sum_{ijk}\mathcal{X}_{ijk}^2}$.

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I. INTRODUCTION

HYPERSPECTRAL images (HSIs) have a high spectral resolution using hundreds of continuous narrow-band ranging from infrared to ultraviolet [1], [2], [3], and can accurately describe the reflection spectrum of ground objects, thereby significantly improving the ability to classify and identify ground objects. The superiority of HSI leads to various applications, such as classification [4], [5], [6] and target detection [7], [8], [9]. Specifically, according to whether there is prior information about the target, hyperspectral target detection can be divided into two categories: matched target detection and anomaly detection. This article discusses the latter, a critical task that can be used in various fields, such as precision agriculture, mineral identification, and military surveillance [10], [11], [12].

Hyperspectral anomaly detection aims to find pixels or subpixels whose spectral or spatial signatures are very different from the surrounding pixels without any prior information about the anomaly or background. Usually, anomaly pixels are embedded in a local homogeneous background, and the percentage of anomaly pixels in all pixels is tiny. There are many challenges in hyperspectral anomaly detection, such as the lack of priors about anomalies, complex backgrounds, and imbalanced distribution of background and anomaly. Various methods have been proposed to tackle this problem. A classical statistics-based method called Reed–Xiaoli (RX) detector was proposed in [13], which assumes the background pixels obey multivariate Gaussian distribution while anomaly pixels do not. The entire image pixels are used to calculate the mean and covariance of the background distribution. Then, the Mahalanobis distance-based method is adopted to measure the anomaly of the test pixel. A more adaptive local RX was proposed in [14], which adopts a dual sliding window and only utilizes the pixels in this window to calculate the mean and covariance. Based on the detection principle

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of RX, some other improved RX-based methods have been proposed, such as the kernel-RX (KRX) in [15], the cluster KRX in [16], the weighted-RX in [17] and [18], the RX-segmented in [19], and the multiple window-based RX in [20], [21], and [22]. In addition, some collaborative representation and sparse representation methods are also proposed for hyperspectral anomaly detection in [23], [24], [25], and [26], which are based on the concept that the background pixel can be approximately represented by its neighborhood pixels, while the anomaly cannot. With the development of deep learning theory, some deep-learning-based methods also have been proposed for hyperspectral anomaly detection, such as the RGAE in [27], the GAED in [28], the auto-AD in [29], and so on.

In recent years, low-rank and sparse matrix decomposition (LSMD) techniques have been widely used for anomaly detection in HSIs [30], [31], [32], [33], [34]. One of the most popular LSMD-based methods is the robust principal component analysis (RPCA) [35], which decomposes an HSI into a low-rank matrix and a sparse anomaly matrix [30]. However, the RPCA does not consider the effect of noise, which may increase the false alarm rate (FAR). Considering the noise influence, a method called LSMAD was proposed in [31], in which an additional noise matrix is introduced in the decomposition compared with the RPCA, and the Mahalanobis distance is used to detect the probable anomaly. Furthermore, assuming that the noise follows a mixture of Gaussian distributions, a new method called LMSD-MOG is proposed in [32]. Since there are numerous mixed pixels in an HSI, the assumption that the HSI data lie in a single subspace may be inaccurate. Thus, a low-rank and sparse representation (LRASA)-based method [33] uses a dictionary to construct the multiple subspaces, the background is represented by this dictionary, and the $l_{2,1}$ norm is designed for the sparse anomaly. However, the above-mentioned LSDM-based methods ignore the structure characteristics of the HSI data and the piecewise smoothness of the background, which may cause a detection performance decrease. To preserve the structural characteristics and the piecewise smoothness, a graph and total variation (TV) regularized low-rank representation (GTVLRR) was proposed in [34].

Actually, HSI data can be treated as a three-order tensor with one spectral and two spatial dimensions. Unfortunately, the matrix-based methods mentioned above usually ignore this inherent multidimensional structure, which may degrade their detection performance. Fortunately, some tensor-based hyperspectral anomaly detection methods that decompose the original HSI data into a background tensor and an anomaly tensor have been proposed, e.g., tensor completion-based method [36], tensor decomposition-based methods (Tucker tensor decomposition-based methods in [37] and [38], CANDECOMP/PARAFAC-based method in [39], TPCA in [40], and ETRPCAMD in [41]) and tensor-regression-based method in [42]. The tensor-based methods can preserve the multidimensional structure of the hyperspectral, thus can improve the anomaly detection performance to a certain extent. However, these tensor-based methods do not comprehensively exploit the latent information of the background, such as the combination of the piecewise smoothness and low-rank properties of the background, thus degrading their performance. A tensor-based method called PTA was proposed in [43], which uses the low-rank property of the background existing in the spectral dimension, the smoothness of the background existing in the spatial dimension, and the sparsity of the anomaly to improve the detection performance. However, PTA ignores the low-rank property of the spatial dimension and the smoothness of the spectral dimension existing in the background. To fully exploit the global low-rank property of the background, a low average rank with a TV regularization model for hyperspectral anomaly detection (LARTVAD) was proposed in [44]. In LARTVAD, the background tensor is represented by a fixed dictionary matrix and a low-rank coefficient tensor. The tensor nuclear norm (TNN) is adopted to constrain the low-rank property of this coefficient tensor. However, TNN-based method is the suboptimal solution for the low-rank constraint [45]. Meanwhile, LARTVAD does not use the piecewise smoothness of the background in spectral dimension.

In fact, the low-rank property of the background exists both in the spatial and spectral dimensions, and the piecewise smoothness of the background also exists in the three dimensions, as shown in Fig. 1. Although the low-rank and piecewise smoothness properties of the HSIs in spatial and spectral dimensions have been utilized at the same time in the HSI denoise [46], [47], [48], HSI and multispectral image fusion [49] and so on, no one completely combines these characteristics of the background in the hyperspectral anomaly detection field. Tensor ring (TR) decomposition with TV regularization can efficiently exploit the low-rank and piecewise smoothness of the hyperspectral. He et al. [50], the author proposed a TR decomposition with TV regularization on spatial dimensions for hyperspectral reconstruction. However, the piecewise smoothness of the spectral dimension is ignored in this article. Further, Chen et al. [49] proposed the TR decomposition with factors smoothed model for hyperspectral and multispectral mixing. Since the low-rank and piecewise smoothness of the background exist both in spatial and spectral dimensions, to make full use of the low-rank and piecewise smoothness of the background, inspired by Chen et al. [49], we proposed a TR decomposition with factors TV regularization model for hyperspectral anomaly detection (TRDFTVAD) in this article. The model in this article can be seen as an extended application of TR decomposition with factors TV regularization in hyperspectral anomaly detection. The overall flowchart of the TRDFTVAD is illustrated in Fig. 1. First, the raw HSI data are decomposed into a background tensor and an anomaly tensor. Then, the TR decomposition is adopted to efficiently exploit the low-rank property of the background existing in each dimension. To make full use of the piecewise smoothness of the background existing in spatial and spectral dimensions, we designed TV regularization on each dimension of the background. Further, this TV regularization is transformed into each TR factor tensor by exploiting the relationship between the background tensor and each factor tensor. Next, the $l_{2,1}$ norm regularization is designed on the anomaly tensor to exploit the group sparsity

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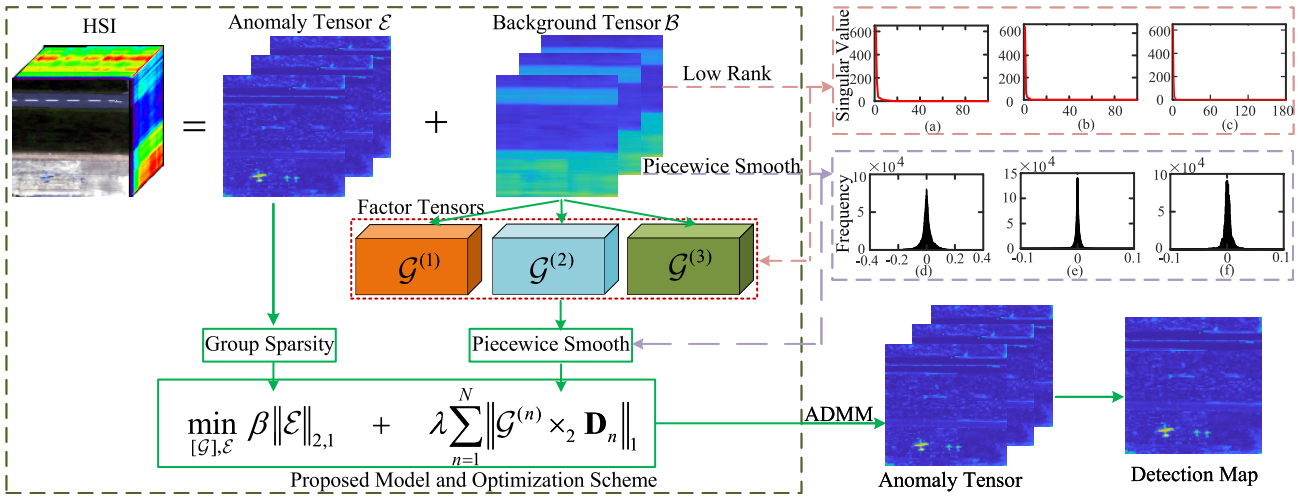


Fig. 1. Flowchart of the proposed TRDFTVAD for hyperspectral anomaly detection. (a)–(c) Eigenvalues of matrices $\mathbf{B}_{(1)}$, $\mathbf{B}_{(2)}$, and $\mathbf{B}_{(3)}$, respectively. (d)–(f) Histograms of the values of $\mathbf{D}_1\mathbf{B}_{(1)}$, $\mathbf{D}_2\mathbf{B}_{(2)}$, and $\mathbf{D}_3\mathbf{B}_{(3)}$, respectively.

of anomaly pixels. Finally, the alternating direction method of multipliers (ADMM) scheme is adopted to update the involved variables with all prior properties taken into account. Compared with existing tensor-based hyperspectral anomaly detection methods, the main contributions of this article are summarized as follows.

1) To the best of our knowledge, this is the first time that simultaneously exploits the low-rank property and piecewise smoothness of the background existing in both spatial and spectral dimensions in hyperspectral anomaly detection under a tensor model.
2) A TR decomposition with factors TV regularization framework is introduced to exploit the low-rank properties and piecewise smoothness of the background in both spatial and spectral dimensions. By fully utilizing the prior information on all dimensions of the background tensor, more accurate anomaly detection results can be obtained.
3) An efficient ADMM-based approach is proposed to solve the proposed TRDFTVAD model. Some experiments are conducted on the real hyperspectral to demonstrate the effectiveness of the proposed method.

The remainder of this article is organized as follows. Some definitions and notations are introduced in Section II. In Section III, the model of the proposed method is given, and the solution process based on ADMM is introduced. The computational complexity of the proposed method is also analyzed. In Section IV, simulation results on several real datasets are given to verify the effectiveness of the proposed method. Finally, a summary is concluded in Section V.

II. Tensor Nation

We list some symbols used in this article and their corresponding meanings in Nomenclature. The followings are some definitions of tensors.

Definition 1 (Mode-k Unfolding): For an Nth-order tensor $\mathcal{X} \in \mathbb{C}^{I_1\times I_2\times\cdots\times I_N}$, its mode-$k$ unfolding $\mathbf{X}_{(k)}$ is an $I_k \times \prod_{n\neq k} I_n$ matrix, which satisfies that $\mathbf{X}_{(k)}(i_k, j)$ is mapped by $\mathcal{X}(i_1, i_2, \ldots, i_N)$, where

$$j = 1 + \sum_{(n=1,n\neq k)}^{N} (i_n - 1)\cdot J_n \tag{1}$$

with $J_n = \sum_{m=1,m\neq n}^{n-1} I_m$. The corresponding operator and inverse operator are denoted as $\mathbf{X}_{(k)} = \text{unfold}_k(\mathcal{X})$ and $\mathcal{X} = \text{fold}_k(\mathbf{X}_{(k)})$, respectively.

Definition 2 (Mode-n Tensor-Matrix Product): The mode-n product of a tensor $\mathcal{X} \in \mathbb{C}^{I_1\times I_2\times\cdots\times I_N}$ with a matrix $\mathbf{A} \in \mathbb{C}^{J_n\times I_n}$ is denoted by $\mathcal{X}_1 = \mathcal{X} \times_n \mathbf{A}$, where $\mathcal{X}_1 \in \mathbb{C}^{I_1\times\cdots I_{n-1}\times J_n\times I_{n+1}\times\cdots I_N}$

$$(\mathcal{X} \times_k \mathbf{A})_{i_1,\ldots,i_{k-1},j,i_{k+1},\ldots,i_N} = \sum_{i_k=1}^{I_k} x_{i_1,i_2,\ldots,i_N}\cdot a_{j,i_k}.$$

Moreover, the mode-n product satisfies the following characteristic:

$$\mathcal{X}_1 = \mathcal{X} \times_n \mathbf{A} \iff \mathbf{X}_{1(n)} = \mathbf{A}\mathbf{X}_{(n)}.$$

Definition 3 (Reversed Mode-n Unfolding [51]): The reversed mode-n unfolding a tensor $\mathcal{X} \in \mathbb{C}^{I_1\times I_2\times\cdots\times I_N}$ is denoted by $\mathbf{X}_{\langle n\rangle} \in \mathbb{C}^{I_n\times(I_{n+1},\ldots,I_N I_1,\ldots,I_{n-1})}$, which satisfies that $\mathbf{X}_{\langle n\rangle}(i_n, j)$ is mapped by $\mathcal{X}(i_1, i_2, \ldots, i_N)$, where

$$j = i_{n+1} + \sum_{s=n+2}^{N}\left((i_s - 1)\prod_{m=n+1}^{s-1}\right) + \sum_{s=1}^{n-1}\left((i_s - 1)\prod_{m=1}^{s-1}\right)\prod_{l=n+1}^{N} I_l.$$

Definition 4 (TR Decomposition [50]): TR decomposition decomposes a tensor into a series of third-order factor tensor $\mathcal{G} = \{\mathcal{G}^{(1)}, \mathcal{G}^{(2)}, \ldots, \mathcal{G}^{(N)}\}$, where $\mathcal{G}^{(n)} \in \mathbb{C}^{r_n\times I_n\times r_{n+1}}$. For an Nth-order tensor $\mathcal{X} \in \mathbb{C}^{I_1\times I_2\times\cdots I_N}$, its TR decomposition is defined as

$$\mathcal{X}(i_1, i_2, \ldots, i_N) = \text{Tr}\big(\mathbf{G}^{(1)}(i_1)\mathbf{G}^{(2)}(i_2), \ldots, \mathbf{G}^{(N)}(i_N)\big)$$

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$$= \mathrm{Tr}\left(\prod_{n=1}^{N} \mathbf{G}^{(n)}(i_n)\right)$$

where $\mathbf{G}^{(n)}(i_n) \in \mathbb{C}^{r_n \times r_{n+1}}$ represents the i_nth lateral slice matrix of factor tensor $\mathcal{G}^{(n)}$ and $\mathrm{Tr}(\cdot)$ denotes the matrix trace operation. Since trace operation is defined for square matrix, TR decomposition sets $r_1 = r_{N+1}$, and $\mathbf{r} = [r_1, r_2, \ldots, r_N]$ is the TR-rank of tensor $\mathcal{X}$. In this article, TR decomposition is simply denoted as $\mathcal{X} = \Phi([\mathcal{G}])$. Give a TR decomposition $\mathcal{X} = \Phi(\mathcal{G}^{(1)}, \mathcal{G}^{(2)}, \ldots, \mathcal{G}^{(N)})$, if $\overleftarrow{\mathcal{X}}^k \in \mathbb{C}^{I_{k+1} \times \cdots I_N \times I_1 \times \cdots I_k}$ denotes the cyclic permutation of $\mathcal{X}$ by k, then $\overleftarrow{\mathcal{X}}^k = \Phi(\mathcal{G}^{(k)}, \ldots, \mathcal{G}^{(N)}, \mathcal{G}^{(1)} \ldots, \mathcal{G}^{(k-1)})$. Thus, each factor tensor $\mathcal{G}^{(n)}$ can be shifted to the first position. Based on this property, the matrix representation of the TR decomposition can be formulated as

$$\mathbf{X}_{\langle k \rangle} = \mathbf{G}_{\langle 2 \rangle}^{(k)} \left(\mathbf{G}_{\langle 2 \rangle}^{(\neq k)}\right)^{\mathrm{T}}$$

which is the based idea to update each factor tensor during TR decomposition. The other efficient property is that, if the TR decomposition of $\mathcal{X}$ is $\mathcal{X} = \Phi(\mathcal{G}^{(1)}, \mathcal{G}^{(2)}, \ldots, \mathcal{G}^{(N)})$, then the mode-$k$ multiplication of a matrix $\mathbf{A}$ can be formulated as

$$\mathcal{X} \times_k \mathbf{A} = \Phi\left(\mathcal{G}^{(1)}, \ldots, \mathcal{G}^{(k)} \times_2 \mathbf{A}, \ldots, \mathcal{G}^{(N)}\right). \tag{2}$$

III. Proposed Method

A. Problem Formulation

Generally speaking, HSI data is a 3-D cube which can be expressed as $\mathcal{Y} \in \mathbb{C}^{w \times h \times b}$, where w and h denote the size of spatial width and height, and b is the number of spectral bands. In the existing tensor-based methods, it is generally assumed that the original cube $\mathcal{Y}$ can be decomposed into a background tensor $\mathcal{X} \in \mathbb{C}^{w \times h \times b}$ and an anomaly tensor $\mathcal{E} \in \mathbb{C}^{w \times h \times b}$. Therefore, a more intuitive model can be expressed as

$$\mathcal{Y} = \mathcal{X} + \mathcal{E}. \tag{3}$$

For hyperspectral anomaly detection, the critical problem is isolating the anomaly pixels from the given raw tensor without any prior information about the anomaly pixels. Therefore, it is essential to use the intrinsic structural information of $\mathcal{X}$ and $\mathcal{E}$ for anomaly detection. On the one hand, the pixels in each band of the background tensor are highly correlated [23], [24], [25], [26], which means that the low-rank property of the background exists in the spectral dimension. This can be reflected by Fig. 1(c). On the other hand, each band of the HSI data $\mathcal{Y}$ can be regarded as an $w \times h$ image composed of the background and anomaly images. The background part contains redundant spatial information for each band image and can be regarded as a low-rank principal component [44]. Thus, the low-rank property of the background exists in the two spatial dimensions. This also can be reflected by Fig. 1(a) and (b). Through the above analysis, we know that the low-rank property of the background exists in both spatial and spectral dimensions. We will fully exploit this property to design the hyperspectral anomaly detection model in the following.

TR decomposition is a novel tensor decomposition technology that decomposes the high-order tensor into a series of three-order factor tensors. This decomposition can explore the low-rank property of the HSI in each dimension, which has been adopted in the hyperspectral reconstruction [50] and HSI and multispectral image fusion [49] and has achieved a good performance. Motivated by this concept, we adopted TR decomposition to explore the low-rank property of the background existing in the spatial and spectral dimensions. Similarly, $l_{2,1}$ regularization is also adopted to exploit the sparsity of the anomaly. Thus, the hyperspectral anomaly detection model based on TR decomposition can be constructed as

$$\begin{aligned} &\min_{[\mathcal{G}],\mathcal{E}} \ \|\mathcal{E}\|_{2,1} \\ &\text{s.t. } \mathcal{Y} = \Psi([\mathcal{G}]) + \mathcal{E} \end{aligned} \tag{4}$$

where $[\mathcal{G}] = \{\mathcal{G}^{(1)}, \mathcal{G}^{(3)}, \mathcal{G}^{(3)}\}$, and $\mathcal{G}^{(1)} \in \mathbb{C}^{r_1 \times w \times r_2}$, $\mathcal{G}^{(2)} \in \mathbb{C}^{r_2 \times h \times r_3}$, $\mathcal{G}^{(3)} \in \mathbb{C}^{r_3 \times b \times r_1}$ are the factor tensors, with r_1, r_2 and r_3 being the TR-rank.

In addition to the low-rank property of the background and the sparsity of anomalies, the piecewise smoothness of the background is also very useful for anomaly detection. From Fig. 1(d)–(f), we can see that the piecewise smoothness of the background exists in both spatial and spectral dimensions. Existing tensor-based methods only design regularization on the spatial dimensions to exploit the piecewise smoothness, ignoring the spectral dimension. To fully exploit the piecewise smoothness of the background existing both in spatial and spectral dimensions, we designed TV regularization on each dimension of the background, i.e.,

$$\|\mathcal{B}\|_{\mathrm{TV}} = \sum_{n=1}^{n=3} \left\|\mathbf{D}_n \mathbf{B}_{\langle n \rangle}\right\|_1 \tag{5}$$

where $\mathbf{D}_n$ is the first-order difference matrix. According to $\mathbf{B}_{\langle n \rangle} = \mathbf{G}_{\langle 2 \rangle}^{(n)} (\mathbf{G}_{\langle 2 \rangle}^{(\neq n),k})^{\mathrm{T}}$, thus we have

$$\mathbf{G}_{\langle 2 \rangle}^{(n)} = \mathbf{B}_{\langle n \rangle} \left(\left(\mathbf{G}_{\langle 2 \rangle}^{(\neq n),k}\right)^{\mathrm{T}}\right)^{\dagger} \tag{6}$$

where $((\mathbf{G}_{\langle 2 \rangle}^{(\neq n),k})^{\mathrm{T}})^{\dagger}$ is the Moore-Penrose of $(\mathbf{G}_{\langle 2 \rangle}^{(\neq n),k})^{\mathrm{T}}$. We can see from (6) that the columns of $\mathbf{G}_{\langle 2 \rangle}^{(n)}$ can be regarded as a linear combination of all columns of $\mathbf{B}_{\langle n \rangle}$. Since the background exists the piecewise smoothness in each dimension, thus the columns of $\mathbf{G}_{\langle 2 \rangle}^{(n)}$ inherent the piecewise smoothness of the columns of $\mathbf{B}_{\langle n \rangle}$. So, we have

$$\|\mathcal{B}\|_{\mathrm{TV}} \Leftrightarrow \sum_{n=1}^{n=3} \left\|\mathbf{D}_n \mathbf{G}_{\langle 2 \rangle}^{(n)}\right\|_1 = \sum_{n=1}^{n=3} \left\|\mathcal{G}^{(n)} \times_2 \mathbf{D}_n\right\|_1. \tag{7}$$

Combined with the above analysis, to fully exploit piecewise smoothness of the background existing both in spatial and spectral dimensions under the framework of TR decomposition simultaneously, our TRDFTVAD can be constructed as

$$\begin{aligned} &\min_{[\mathcal{G}],\mathcal{E}} \ \lambda \sum_{n=1}^{n=3} \left\|\mathcal{G}^{(n)} \times_2 \mathbf{D}_n\right\|_1 + \beta \|\mathcal{E}\|_{2,1} \\ &\text{s.t. } \mathcal{Y} = \Psi([\mathcal{G}]) + \mathcal{E} \end{aligned} \tag{8}$$

where λ and β are two positive weight parameters to control the TV regularization and the $l_{2,1}$ regularization, respectively. Under the framework of TR decomposition, the low-rank

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background tensor is mainly approximated by updating the factor tensors, so the different prior properties of the background, except the low-rank property, should learn through the updating of the factor tensors. In the proposed TRDFTVAD model, the piecewise smoothness property of the background tensor is exploited by designing TV regularization on each TR factor tensor. Using this prior property to guide the updating of the factor tensor makes the final estimated background tensor more approximate to the actual background tensor. All in all, the low-rank and piecewise smoothness properties of the background existing in both spatial and spectral dimensions and the group sparsity of the anomaly are all taken into account under the TR decomposition framework, making the detection result more accurate.

B. Optimization

To solve the model in (8), we apply the ADMM, which is efficient and widely used. Moreover, because the variables of TRDFTVAD are interdependent, we impose some auxiliary variables to simplify the optimization. Thus, the proposed TRDFTVAD model can be rewritten as

$$\min_{[\mathcal{S}],[\mathcal{G}],\mathcal{E}} \lambda\sum_{n=1}^{n=3}\left\|\mathcal{S}^{(n)}\right\|_1+\beta\|\mathcal{E}\|_{2,1} \quad \text{s.t.} \begin{cases}\mathcal{S}^{(n)}=\mathcal{G}^{(n)}\times_2\mathbf{D}_n\\ \mathcal{Y}=\Psi([\mathcal{G}])+\mathcal{E}\end{cases} \tag{9}$$

where $\mathcal{S}^{(n)}$, $(n=1,2,3)$ are the auxiliary variables. By merging the equal constraints of the auxiliary variables into the Lagrangian equation, the augmented Lagrangian function of the TRDFTVAD model can be rewritten as

$$\begin{aligned}L([\mathcal{G}],\mathcal{E},[\mathcal{S}],[\mathcal{P}],\mathcal{R}) &= \lambda\sum_{n=1}^{n=3}\left\|\mathcal{S}^{(n)}\right\|_1+\beta\|\mathcal{E}\|_{2,1}\\ &+\sum_{n=1}^{n=3}\Big(+\frac{\mu}{2}\left\|\mathcal{G}^{(n)}\times_2\mathbf{D}_n-\mathcal{S}^{(n)}\right\|_{\mathrm{F}}^2\\ &\qquad+\left\langle\mathcal{P}^{(n)},\mathcal{G}^{(n)}\times_2\mathbf{D}_n-\mathcal{S}^{(n)}\right\rangle\Big)\\ &+\frac{\mu}{2}\|\mathcal{Y}-\Psi([\mathcal{G}])-\mathcal{E}\|_{\mathrm{F}}^2\\ &+\langle\mathcal{R},\mathcal{Y}-\Psi([\mathcal{G}])-\mathcal{E}\rangle\end{aligned} \tag{10}$$

where $[\mathcal{P}]=\{\mathcal{P}^{(n)}\}_{n=1}^{n=3}$ and $\mathcal{R}$ are the Lagrangian multipliers, and $\mu>0$ is a positive penalty parameter. So far, for $n=1,2,3$, $\mathcal{G}^{(n)}$, $\mathcal{P}^{(n)}$, $\mathcal{S}^{(n)}$, $\mathcal{R}$ and $\mathcal{E}$ are each independent. Thus, we can update one of them with another fixed by the following scheme.

1) *Update $\mathcal{G}^{(n)}(n=1,2,3)$:* Using (10), the augmented Lagrangian function with respect to $\mathcal{G}^{(n)}$ can be constructed as

$$\begin{aligned}\mathcal{G}^{(n),k+1} &= \min_{\mathcal{G}^{(n)}}\frac{\mu^k}{2}\left\|\mathcal{G}^{(n)}\times_2\mathbf{D}_n-\mathcal{S}^{(n),k}\right\|_{\mathrm{F}}^2\\ &+\left\langle\mathcal{P}^{(n),k},\mathcal{G}^{(n)}\times_2\mathbf{D}_n-\mathcal{S}^{(n),k}\right\rangle\\ &+\frac{\mu^k}{2}\left\|\mathcal{Y}-\Psi([\mathcal{G}])-\mathcal{E}^k\right\|_{\mathrm{F}}^2\\ &+\left\langle\mathcal{R}^k,\mathcal{Y}-\Psi([\mathcal{G}])-\mathcal{E}^k\right\rangle\\ &=\min_{\mathcal{G}^{(n)}}\frac{\mu^k}{2}\left\|\mathcal{G}^{(n)}\times_2\mathbf{D}_n-\mathcal{S}^{(n),k}+\frac{\mathcal{P}^{(n),k}}{\mu^k}\right\|_{\mathrm{F}}^2\\ &+\frac{\mu^k}{2}\left\|\mathcal{Y}-\Psi([\mathcal{G}])-\mathcal{E}^k+\frac{\mathcal{R}^k}{\mu^k}\right\|_{\mathrm{F}}^2.\end{aligned} \tag{11}$$

Based on the matrix representation of TR decomposition and let $\mathbf{P}_1^k=(\mathbf{G}_{\langle 2\rangle}^{(\neq n),k})^{\mathrm{T}}$. Then, the problem of (11) can be converted into the following optimization problem:

$$\begin{aligned}\mathbf{G}_{(2)}^{(n),k+1} &= \min_{\mathbf{G}_{(2)}^{(n)}}\frac{\mu^k}{2}\left\|\mathbf{D}_n\mathbf{G}_{(2)}^{(n)}-\mathbf{S}_{(2)}^{(n),k}+\frac{\mathbf{P}_{(2)}^{(n),k}}{\mu^k}\right\|_{\mathrm{F}}^2\\ &+\frac{\mu}{2}\left\|\mathbf{Y}_{\langle n\rangle}-\mathbf{G}_{(2)}^{(n)}\mathbf{P}_1^k-\mathcal{E}_{\langle n\rangle}^k+\frac{\mathbf{R}_{\langle n\rangle}^k}{\mu^k}\right\|_{\mathrm{F}}^2\end{aligned} \tag{12}$$

which is a quadratic problem, and the closed-form solution of $\mathbf{G}_{(2)}^{(n)}$ can be obtained by solving the following linear system:

$$\begin{aligned}&\mathbf{D}_n^{\mathrm{T}}\mathbf{D}_n\mathbf{G}_{(2)}^{(n)}+\mathbf{G}_{(2)}^{(n)}\mathbf{P}_1^k(\mathbf{P}_1^k)^{\mathrm{T}}\\ &=\mathbf{D}_n^{\mathrm{T}}\left(\mathbf{S}_{(2)}^{(n),k}-\frac{\mathbf{P}_{(2)}^{(n),k}}{\mu^k}\right)\\ &\quad+\left(\mathbf{Y}_{\langle n\rangle}-\mathcal{E}_{\langle n\rangle}^k+\frac{\mathbf{R}_{\langle n\rangle}^k}{\mu^k}\right)(\mathbf{P}_1^k)^{\mathrm{T}}.\end{aligned} \tag{13}$$

The above formulation is a classical Sylvester matrix equation, which can be efficiently solved according to [52]. Then, the updated factor $\mathcal{G}^{(n)}$ is obtained by folding the solution $\mathbf{G}_{(2)}^{(n),k+1}$, i.e., $\mathcal{G}^{(n),k+1}=\mathrm{fold}_2(\mathbf{G}_{(2)}^{(n),k+1})$.

2) *Update $\mathcal{S}^{(n)}$:* For $n=1,2,3$, the augmented Lagrangian function with respect to $\mathcal{S}^{(n)}$ can be constructed as

$$\begin{aligned}&\mathcal{S}^{(n),k+1}\\ &=\min_{\mathcal{S}^{(n)}}\lambda\left\|\mathcal{S}^{(n)}\right\|_1+\frac{\mu^k}{2}\left\|\mathcal{S}^{(n)}-\mathcal{G}^{(n),k+1}\times_2\mathbf{D}_n\right\|_{\mathrm{F}}^2\\ &\quad+\left\langle\mathcal{P}^{(n),k},\mathcal{S}^{(n)}-\mathcal{G}^{(n),k+1}\times_2\mathbf{D}_n\right\rangle\\ &=\min_{\mathcal{S}^{(n)}}\lambda\left\|\mathcal{S}^{(n)}\right\|_1\\ &\quad+\frac{\mu^k}{2}\left\|\mathcal{S}^{(n)}-\mathcal{G}^{(n),k+1}\times_2\mathbf{D}_n+\frac{\mathcal{P}^{(n),k}}{\mu^k}\right\|_{\mathrm{F}}^2.\end{aligned} \tag{14}$$

Define

$$\tilde{\mathcal{S}}^{(n),k}=\mathcal{G}^{(n),k+1}\times_2\mathbf{D}_n-\frac{1}{\mu^k}\mathcal{P}^{(n),k} \tag{15}$$

and then, (14) can be solved by the soft thresholding

$$\mathcal{S}^{(n),k+1}(i_w,i_h,i_b)=\mathrm{soft}_1\left(\tilde{\mathcal{S}}^{(n),k}(i_w,i_h,i_b),\frac{\lambda}{\mu^k}\right) \tag{16}$$

where $(i_w=1,\ldots,w)$, $(i_h=1,\ldots,h)$, $(i_b=1,\ldots,b)$ and

$$\mathrm{soft}_1(x,a)=\mathrm{sign}(x)\max(|x|-a,0). \tag{17}$$

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3) *Update $\mathcal{E}$:* The augmented Lagrangian function with respect to $\mathcal{E}$ can be expressed as

$$\begin{aligned}\mathcal{E}^{k+1} &= \min_{\mathcal{E}} \beta\|\mathcal{E}\|_{2,1} + \frac{\mu^k}{2}\|\mathcal{Y} - \Psi([\mathcal{G}]) - \mathcal{E}\|_{\mathrm{F}}^2 \\ &\quad + \left\langle \mathcal{R}^k, \mathcal{Y} - \Psi([\mathcal{G}]) - \mathcal{E}\right\rangle \\ &= \min_{\mathcal{E}} \beta\|\mathcal{E}\|_{2,1} \\ &\quad + \frac{\mu^k}{2}\left\|\mathcal{Y} - \Psi([\mathcal{G}]) - \mathcal{E} + \frac{\mathcal{R}^k}{\mu^k}\right\|_{\mathrm{F}}^2. \end{aligned} \tag{18}$$

Define

$$\tilde{\mathcal{E}}^k = \mathcal{Y} - \Psi([\mathcal{G}]) + \frac{1}{\mu^k}\mathcal{R}^k$$

and then, (18) can be solved by

$$\mathcal{E}^{k+1}(i_w, i_h, :) = \mathrm{soft}_{2,1}\left(\tilde{\mathcal{E}}^k(i_w, i_h, :), \frac{\beta}{\mu^k}\right) \tag{19}$$

where

$$\mathrm{soft}_{2,1}(\mathbf{x}, a) = \max\left(1 - \frac{a}{\|\mathbf{x}\|_2}, 0\right) \cdot \mathbf{x}. \tag{20}$$

4) *Update Multipliers:* For $n = (1, 2, 3)$, the multipliers and the penalty parameter can be updated as follows:

$$\begin{cases} \mathcal{P}^{(n),k+1} = \mathcal{P}^{(n),k} + \mu^k\left(\mathcal{G}^{(n),k+1} \times_2 \mathbf{D}_n - \mathcal{S}^{(n),k+1}\right) \\ \mathcal{R}^{k+1} = \mathcal{R}^k + \mu^k\left(\mathcal{Y} - \Phi([\mathcal{G}]) - \mathcal{E}^{k+1}\right) \\ \mu^{k+1} = \min\left(\rho\mu^k, \mu_{\max}\right) \end{cases} \tag{21}$$

where $\rho > 1$ is a constant to reduce the impact of the initialization of μ, and $\mu_{\max}$ is the upper bound of μ.

Algorithm 1 Proposed TRDFTVAD Algorithm

Input: Tensor data $\mathcal{Y}$, parameters λ, β, TR rank r_1, r_2, r_3.
Initialize: Set the variables $\mathcal{E}$, $\mathcal{S}^{(n)}$, $\mathcal{P}^{(n)}$, $\mathcal{R}$ as zero tensors, random sample $\mathcal{G}^{(n)}$by distribution $N \sim (0, 1)$, and let $k = 1$, $k_{max} = 100$, $\rho = 1.2$, $\mu_0 = 0.01$, $\mu_{\max} = 10^9$, $\varepsilon_0 = 10^{-6}$.
While ($\varepsilon > \varepsilon_0$ or $k < k_{\max}$) **do**
1) Update $\mathcal{G}^{(n)}$ according to eq(13).
2) Update $\mathcal{S}^{(n)}$ by eq(16).
3) Update $\mathcal{E}$ by eq(19).
4) Update $\mathcal{P}^{(n)}$, $\mathcal{Q}^{(n)}$, and μ by eq(21).
6) $\varepsilon = \|\mathcal{B}^k - \mathcal{B}^{k-1}\|_{\mathrm{F}}/\|\mathcal{B}^{k-1}\|_{\mathrm{F}}$.
7) $k = k + 1$.
End while
Output $\mathcal{E}$

C. Complexity Analysis

We analysis the computational complexity of the proposed method by solving an HSI data $\mathcal{Y} \in \mathbb{C}^{w\times h\times b}$. As shown in Algorithm 1, the main computational cost in each iteration relies on the updating of $\mathcal{G}_n$, $\mathcal{S}_n$, $\mathcal{E}$, and $\mathcal{P}^{(n)}$. Updating $\mathcal{G}^{(1)}$, $\mathcal{G}^{(2)}$, and $\mathcal{G}^{(3)}$ needs $O(whbr_1r_2 + r_1^3r_2^3 + r_1r_2w\log(w) + wr_1^2r_2^2)$, $O(whbr_2r_3 + r_2^3r_3^3 + r_2r_3h\log(h) + hr_2^2r_3^2)$, and $O(whbr_3r_1 + r_3^3r_1^3 + r_3r_1b\log(b) + br_3^2r_1^2)$, respectively, since each process involves SVD, 1-D FFT, and several matrix multiplications; Updating $\mathcal{S}^{(1)}$, $\mathcal{S}^{(2)}$, and $\mathcal{S}^{(3)}$ needs $\mathcal{O}(w^2r_1r_2)$, $\mathcal{O}(h^2r_2r_3)$, and $\mathcal{O}(b^2r_3r_1)$, respectively, since each process only involves matrix multiplication; Updating $\mathcal{E}$ needs $\mathcal{O}(r_1r_2r_3wh + r_3r_1^2whb + r_1^2whb)$, since this process only involves matrix multiplication; Updating $\mathcal{P}^{(1)}$, $\mathcal{P}^{(2)}$, and $\mathcal{P}^{(3)}$ needs $\mathcal{O}(w^2r_1r_2)$, $\mathcal{O}(h^2r_2r_3)$, and $\mathcal{O}(b^2r_3r_1)$, respectively, since each process only involves matrix multiplication.

IV. Simulation Results

In this section, we exploit six real hyperspectral datasets to evaluate the performance of the proposed hyperspectral anomaly detection algorithm. A detailed description of these six datasets is given below. Several classical methods are introduced as comparisons to illustrate the effectiveness of the proposed method. All the experiments are carried out in MATLAB R2022a on a personal computer with an Intel Core i9-12900K 3.20-GHz central processing unit, with 64-GB memory and 64-bit Windows 10.

A. Hyperspectral Datasets

The first dataset is the HYDICE dataset captured by the Hyperspectral Digital Imagery Collection Experiment HYDICE), which covers an urban area with a relatively complex background for small target detection.

1) The first dataset was captured by the Hyperspectral Digital Imagery Collection Experiment (HYDICE), which covers an urban area with a relatively complex background for small target detection. This image includes 80 × 100 pixels and 175 influential bands. The spatial resolution is 1.56 m, and the spectral resolution is 10 nm. As shown in Fig. 2, several roofs and cars composed of 21 pixels in this image are regarded as anomalies.
2) The second dataset was captured by the SpecTIR hyperspectral airborne Rochester experiment (SHARE). This experimental data contains 180 × 180 pixels and 120 bands with a 5-nm spectral resolution and 1-m ground spatial resolution, respectively. We select an image area of 180 × 180 pixels with 120 bands for evaluation. In this image scene, normal classes of road and vegetation are the background mostly in contrast to the natural background, and anomalies consist of artificial colorful square fabrics.
3) The third dataset is the urban-3 dataset captured by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor with 3.5-m spatial resolution, which contains 100 × 100 pixels and 191 bands. This is a relatively complex object detection scene consisting of various sizes of the anomaly with 52 pixels.
4) The fourth dataset is the urban-5 dataset captured by AVIRIS sensor with the spatial resolution being 7.1 m. This HSI contains 100 × 100 pixels and 205 effective bands. This is a relatively complex anomaly detection scene with many anomalies consisting of various sizes of aircraft with a total of 232 pixels.
5) The fifth dataset is the airport-2 dataset captured by the AVIRIS sensor with the spatial resolution being 7.14 m, which contains 100 × 100 pixels and 205 bands. This scene has a relatively simple background, where two large aircraft are taken as the anomaly with a total of 87 pixels.

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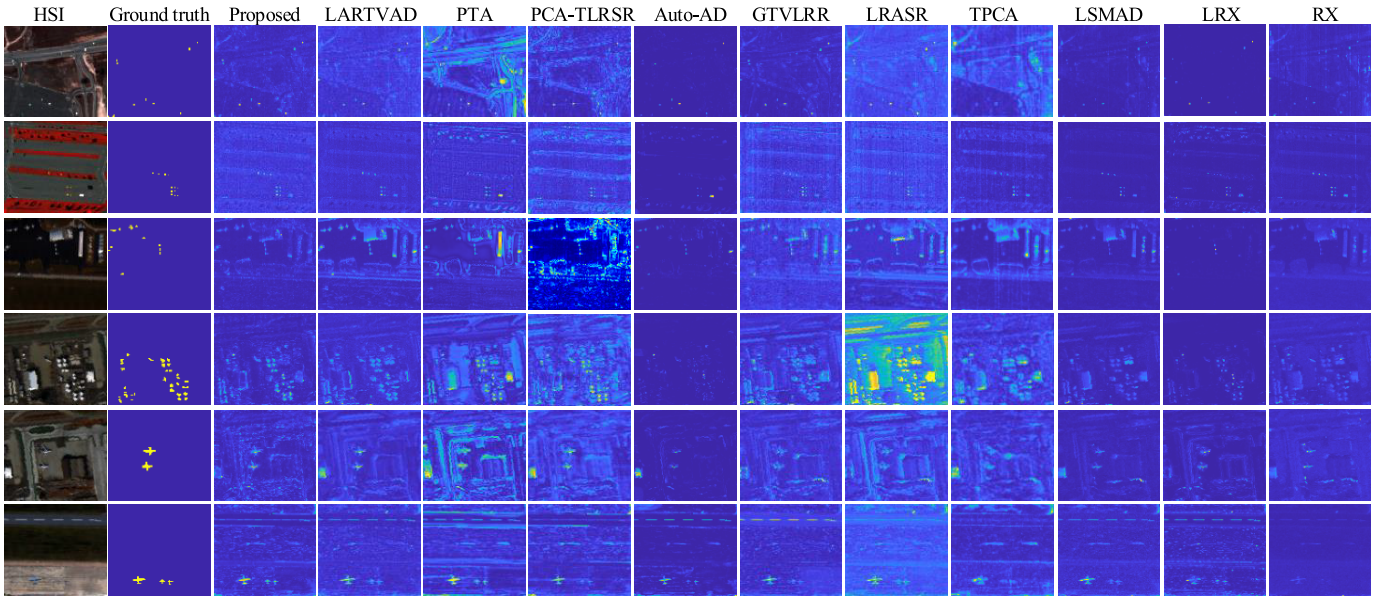


Fig. 2. Anomaly detection map of different methods on the six datasets. (From top to bottom) HYDICE, SpecTIR, urban-3, urban-5, airport-2, and airport-4.

6) The sixth is the airport-4 dataset captured by the AVIRIS sensor with the spatial resolution being 1.56 m, which contains 100 × 100 pixels and 189 bands. This scene has a relatively simple background with a small number of anomalies, where one larger and two small aircraft are taken as the anomaly with a total of 60 pixels.

B. Evaluation Indicators and Comparison methods

The receiver operating characteristic (ROC) curve, a commonly quantitative analysis method to evaluate detection performance in hyperspectral anomaly detection, which indicates the change of detection probability P_D with false alarm probability P_F, is adopted in this paper to evaluate the anomaly detection performance.

1) The ROC curve is a commonly quantitative analysis method to evaluate detection performance in hyperspectral anomaly detection, which indicates the change of detection probability P_D with false alarm probability P_F. P_D and P_F can be calculated by the following equation:

$$P_D = \frac{N_D}{N_A}, \quad P_F = \frac{N_F}{N_B} \tag{22}$$

where N_D represents the number of anomaly pixels in HSI data that are also detected as anomaly pixels correctly. N_A is defined as the total number of anomaly pixels in the HSI data, N_F represents the number of anomaly pixels in HSI data but falsely detected as background pixels during detection, and N_B represents the number of the background pixels in HSI data. In addition, the area under the ROC curve ($\mathrm{AUC}_{(P_D,P_F)}$) was adopted to quantify the detection performance, which idea value is 1.

2) Ten classical methods are introduced as comparisons to demonstrate the effectiveness of the proposed algorithm, including two classical static-based methods called RX [13] and LRX [14], three matrix-based methods, i.e., the LSMAD [31], the LRASR [33], and the GTVLRR [34], four tensor-based methods, i.e., the TPCA [40], the PTA [43], the LARTVAD [44], and the PCA-TLRSR [42], one deep-learning-based method called auto-AD in [29]. For LRX, the parameters of window size (w_{out}, w_{in}) are set as (5, 3). For the background dictionary in the LRASR, the GTVLRR, and the LARTVAD, the number of clusters is set to 15, and 20 pixels are selected in each cluster as atoms. The two parameters λ and β in LRASR, are chosen from $\{0.01, 0.05, 0.1, 0.5, 1, 5, 10\}$. The values of the two parameters λ and β in GTVLRR, are chosen from $\{0.005, 0.05, 0.1, 0.3, 0.5, 0.7, 10\}$. The values of the parameters γ in GTVLRR are chosen from $\{0.005, 0.01, 0.02, 0.05, 0.1, 0.2, 0.5\}$. In the PTA method, the truncated low-rank r is selected from 0 to 20 and the regularization parameter μ is selected from $\{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 10^{0}, 10^{1}, 10^{2}, 10^{3}, 10^{4}\}$. Other parameters of the PTA algorithm remain unchanged as in [43]. The λ and β in LARTVAD, the values of those two parameters are selected from $\{0.01, 0.05, 0.1, 0.5, 1, 5, 10\}$, the number of clusters K is set to 80 for the six datasets. For the PCA-TLRSR, the parameters λ and λ' are chosen from $\{0.001, 0.005, 0.01, 0.05, 0.1, 0.2, 0.3, 0.4, 0.5\}$. As for the auto-AD, the loss threshold σ is set to 1.5×10^{-5}.

C. Experiment Results and Analysis

The detection maps obtained by different methods on the six datasets are shown in Fig. 2 to evaluate the performance of the proposed method visually. For the HYDICE dataset shown in Fig. 2 (first row), the proposed method seems to have the best anomaly detection performance and good background suppression. Meanwhile, the discrimination between the background pixels and the anomaly pixels of the proposed method seems to be the best. The LRX has the best background suppression. In contrast, the PTA has the worst background suppression performance. The discrimination between the background pixels and the anomaly pixels of the TPCA, the RX, the LSMAD, and the PCA-TLRSR seem to be not good. For the SpecTIR

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dataset shown in Fig. 2 (second row), the proposed method, the LARTVAD, and the RX seem to have the same anomaly detection performance. The LSMAD seems to have the best background suppression. Although the auto-AD seems also to have good background suppression, it misses some anomaly pixels. The discrimination between the background pixels and the anomaly pixels of the LSMAD is the best among the compared methods. Meanwhile, the LSMAD seems to have the best background suppression performance among all compared methods. The PCA-LRSR seems to have mediocre anomaly detection performance because it lost some anomaly pixels. The LRX has a bad anomaly detection performance, and the background suppression of this method is also not good. Besides, it seems that the discrimination between the background pixels and the anomaly pixels of the LRASR is the worst. For the urban-3 dataset shown in Fig. 2 (third row), the proposed method, the LARTVAD, the GTVLRR, and the LRASR seem to have the same anomaly detection performance. The PCA-TLRSR seems to have the worst anomaly detection and background suppression performance among all the compared methods. The auto-AD seems to have the best background suppression. The LRX seems to have good background suppression, but it misses some anomaly pixels. The LARTVAD seems to have the best discrimination. Still, the TPCA has the worst anomaly detection performance. For the urban-5 dataset shown in Fig. 2 (fourth row), the proposed method and the LARTVAD seem to have the same detection performance. Still, the discrimination between the background pixels and the anomaly pixels of the proposed method is better than the LARTVAD. The PTA and the PCA-TLRSR seem to have good anomaly detection performance, but it seems to have mediocre background suppression. The auto-AD and the LRX seem to have good background suppression, but they miss a lot of anomaly pixels. Among all compared methods, the TPCA has the worst anomaly detection performance since it misses a lot of anomaly pixels, and the LRASR has the worst background suppression and discrimination. For the airport-2 dataset shown in Fig. 2 (fifth row), the proposed method, the LARTVAD, and the PCA-TLRSR seem to have the same anomaly detection performance, better than other compared methods. But the proposed method has better background suppression than the LARTVAD and the PCA-TLRSR. Besides, the discrimination between the background pixels and the anomaly pixels in the detection map obtained by the proposed method is the highest among all compared methods. The auto-AD seems to have good background suppression, but it misses a lot of anomaly pixels. In contrast, the performance of suppressing the background in the detection map obtained by PTA, LRASR, and TPCA is not good. Besides, the discrimination between the background pixels and the anomaly pixels of the PTA is the worst. For the airport-4 dataset shown in Fig. 2 (sixth row), the proposed method has the best anomaly detection performance among all compared methods. Meanwhile, the discrimination between the background pixels and the anomaly pixels of the two small targets detected by the proposed algorithm is also the best. The auto-AD seems to have good background suppression and a mediocre anomaly detection performance. Although the RX has good background suppression, it has the worst detection performance since it missed most anomaly pixels. The discrimination between the background pixels and the anomaly pixels of the LRASR is also not good. The comparative analysis above shows that the proposed method has good detection and background suppression performance, and the discrimination between background and abnormal pixels is also good on most datasets.

The 2-D and 3-D ROC curves for all compared methods on the six datasets are shown in Figs. 3 and 4, respectively. As we can see from Fig. 3, in urban-5, airport-2, and airport-4 datasets, for almost every probability of the false rate (PF), the detection probability of the proposed method is the highest among all the compared methods. Therefore, the detection performance of the proposed method is the best among all compared methods, especially when the PF is low. For the HYDICE and urban-3 datasets shown in Fig. 3(a) and (c), respectively, the detection probability of the proposed method is highest among all the compared methods when PF is high. For SpecTIR dataset shown in Fig. 3(b), when the PF is relatively low or high, the PD is the highest among all the compared methods. The 3-D ROC shown in Fig. 4 also demonstrates that the performance of the proposed method is relatively more satisfactory than the other compared methods. To further compare the performance of these methods on the six datasets quantitatively, the $\mathrm{AUC}_{(P_D,P_F)}$ scores of all these methods are provided in Table I. The $\mathrm{AUC}_{(P_D,P_F)}$ of the proposed method on the six datasets is the highest among the compared methods, which means that the proposed method has an excellent anomaly detection performance.

In addition, the normalized background-anomaly separation map is adopted to illustrate the performance of different methods for separating anomaly pixels from background pixels. The separation map reflects the statistical characteristics of anomaly pixels and background pixels in the detection results. As shown in Fig. 5, the boxes represent the pixels in the range of 10%–90%, and the horizontal line in each box represents the median value of the pixels. The lines above and below the boxes represent the pixels in the range of 90%–100% and 0%–10%, respectively. We adopt red and blue boxes to denote anomaly and background detection results. The distance between the red and blue boxes represents the separation of background and anomaly. The larger the distance, the better the separation. For the HYDICE dataset shown in Fig. 5(a), the separation performance of the proposed method is close to that of the LARTVAD and LRASR, slightly better than the GTVLRR, but much better than the PTA, the PCA-TLRSR, the auto-AD, the TPCA, the LSMAD, the LRX, and the RX. For the SpecTIR dataset shown in Fig. 5(b), the separation of the proposed method is close to that of the LARTVAD and the RX, slightly better than that of the PTA and the GTVLRR, but much better than other compared methods. For the urban-3 dataset shown in Fig. 5(c), the separation performance of the GTVLRR is the best. The separation of the proposed method is close to that of the LARTVAD and the PTA, much better than other compared methods except GTVLRR. For the urban-5 dataset shown in Fig. 5(d), the separation performance of the proposed method is close to the LARTVAD, much better than other compared methods. For the airport-2 dataset shown in Fig. 5(e), the separation performance of the proposed

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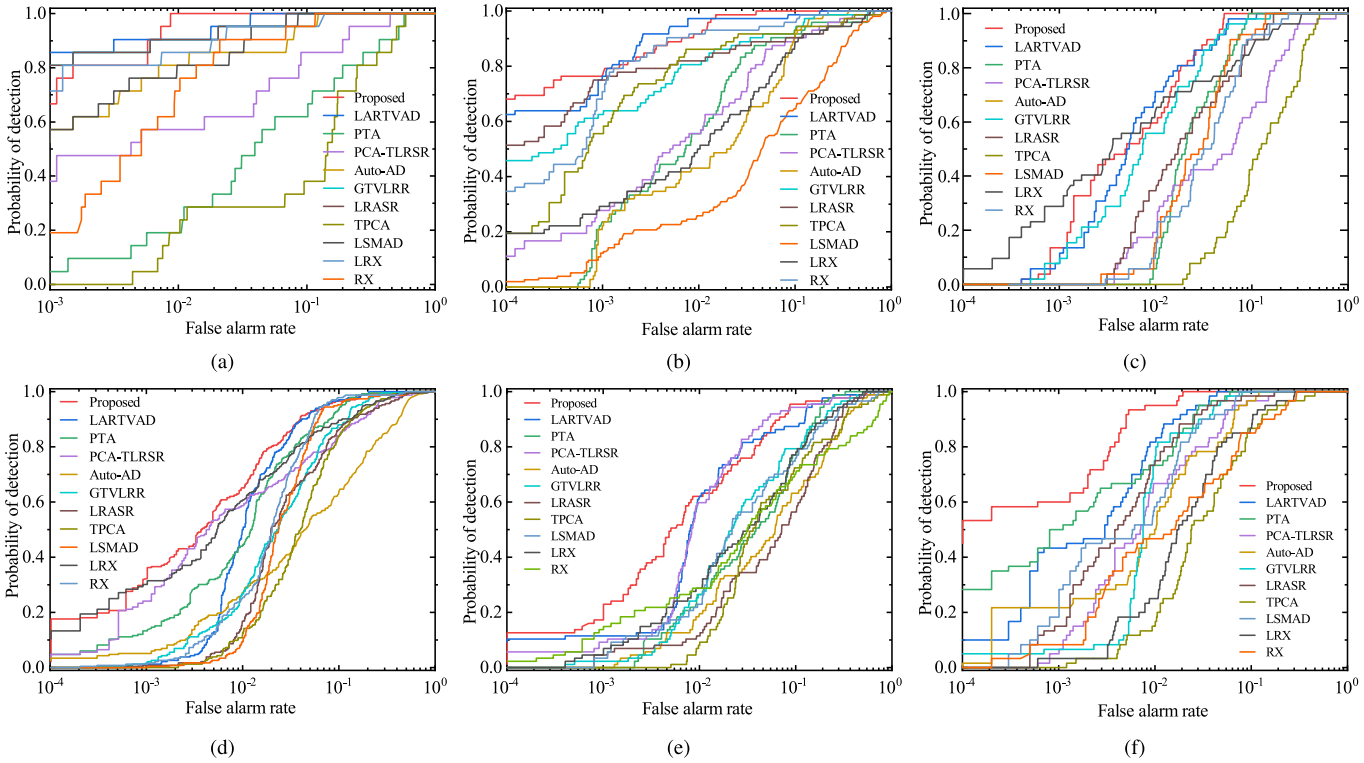


Fig. 3. ROC curves of the proposed method and the compared methods. (a) HYDICE dataset. (b) SpecTIR dataset. (c) Urban-3 dataset. (d) Urban-5 dataset. (e) Airport-2 dataset. (f) Airport-4 dataset.

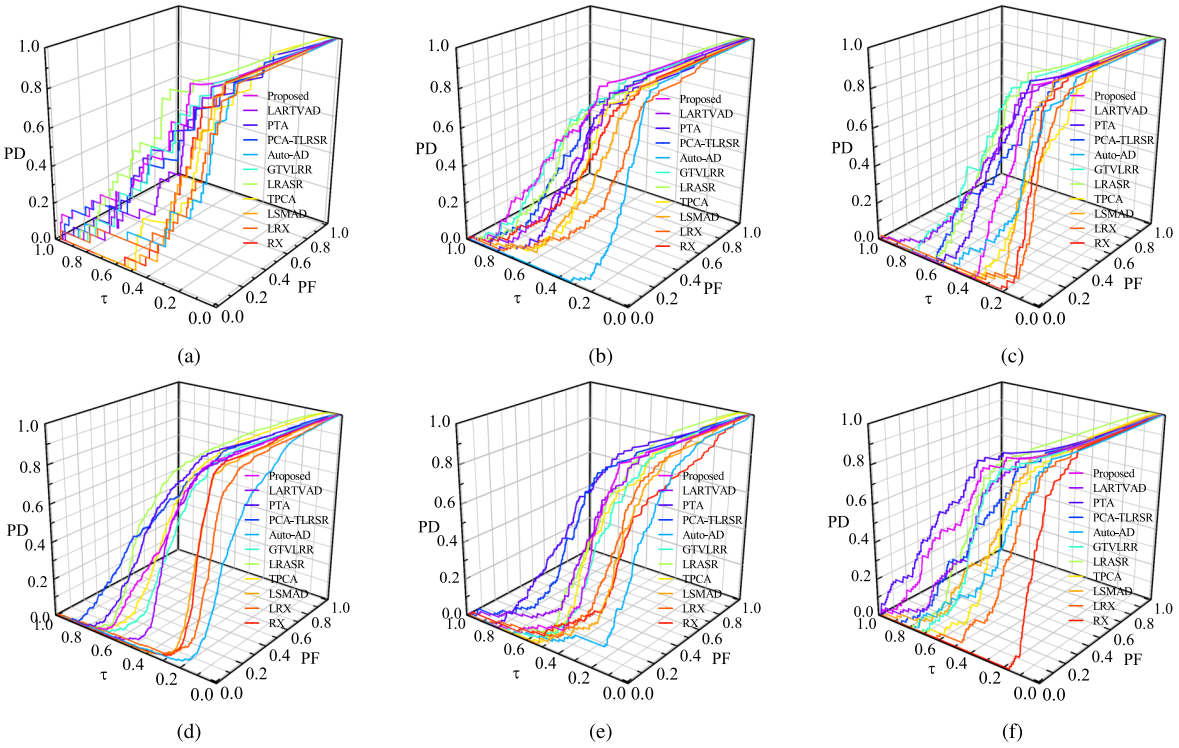


Fig. 4. 3-D-ROC of the proposed method and the compared methods. (a) HYDICE dataset. (b) SpecTIR dataset. (c) Urban-3 dataset. (d) Urban-5 dataset. (e) Airport-2 dataset. (f) Airport-4 dataset.

method and the PCA-TLRSR are much better than other compared methods since there is a clear gap between the background and outlier boxes in the proposed method and the PCA-TLRSR. In contrast, other methods overlap between the background and exception boxes. For the airport-4 dataset shown in Fig. 5(f), the proposed method has the best separation

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TABLE I

$\mathrm{AUC}_{(P_D,P_F)}$ VALUES (%) OF DIFFERENT METHODS FOR THE SIX DATASETS

Dataset	Proposed	LARTVAD	PTA	PCA-TLRSR	Auto-AD	GTVLRR	LARSR	TPCA	LSMAD	LRX	RX
HYDICE	99.8335	99.6956	87.1167	94.3351	98.5307	99.5291	99.6281	82.4196	99.0182	99.1916	98.5536
SpecTIR	99.7990	99.5547	95.9103	95.8900	95.9567	98.0328	96.4915	96.9385	99.6202	94.4694	99.1350
urban-3	98.8001	98.6574	96.9232	89.6766	97.3411	98.2422	96.1721	82.2410	96.3655	96.2858	95.1171
urban-5	98.1537	96.5501	97.3434	94.9432	87.4592	95.5234	94.0123	93.6962	96.1038	96.7704	96.9111
airport-2	97.5774	96.5501	93.6158	96.6649	87.5307	93.5957	88.6835	88.9124	91.8097	94.7061	84.0294
airport-4	99.7746	99.3698	99.2134	98.0286	97.8444	99.5508	98.8531	94.3219	98.6676	95.9757	95.2482

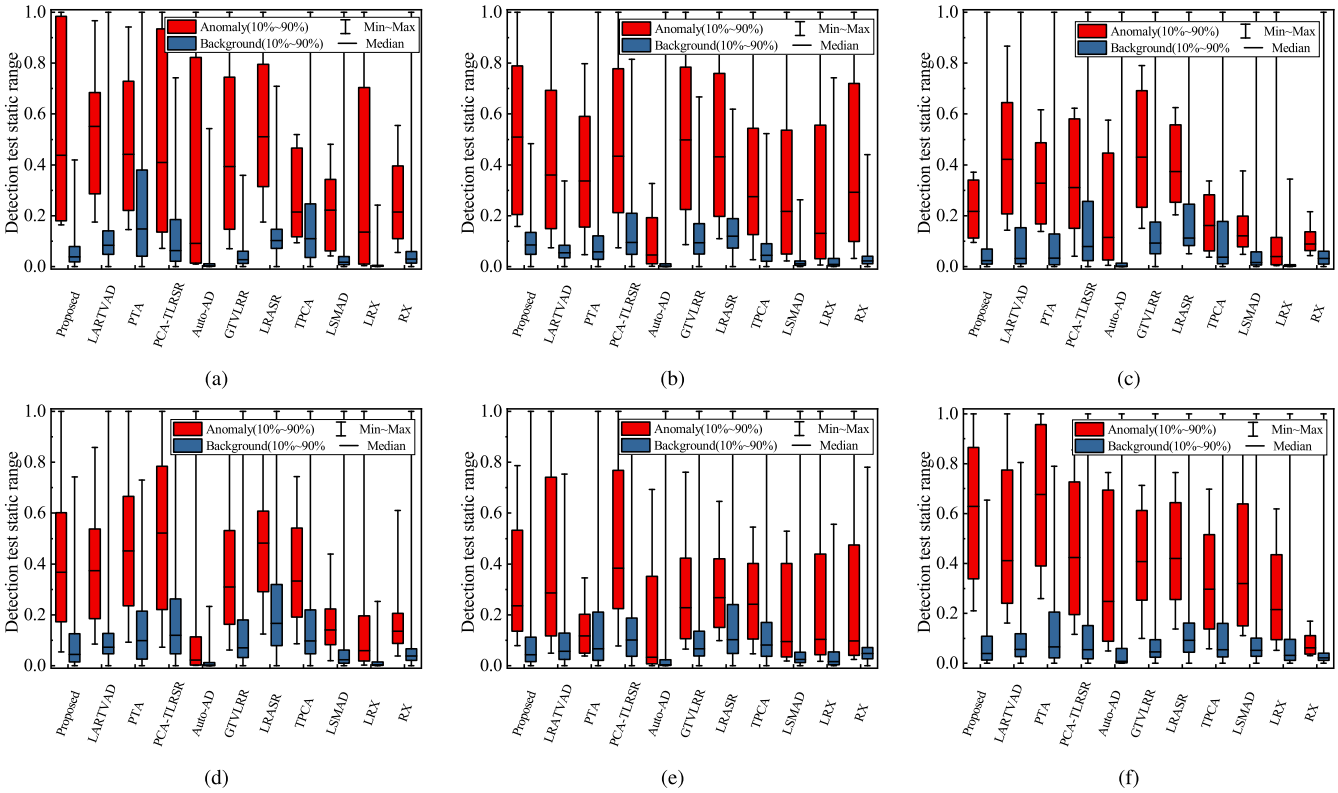


Fig. 5. Background-anomaly separation maps. (a) HYDICE dataset. (b) SpecTIR dataset. (c) Urban-3 dataset. (d) Urban-5 dataset. (e) Airport-2 dataset. (f) Airport-4 dataset.

performance among the comparison methods since the gap between the background and outlier boxes in the proposed method is large than that in other compared methods. Overall, the proposed method shows an excellent background-abnormal separation performance in almost the six datasets.

D. Parameters Setting

There are five regularization parameters λ, β, r_1, r_2, and r_3 and two algorithm parameters μ and ρ that need to be manually set to make the algorithm achieve optimal performance. For the algorithm parameters μ and ρ, we fixed $\mu = 0.01$ and $\rho = 1.2$ in all experiments. Next, we analyze the influence of the TV regularization λ and sparse regularization β parameters on the anomaly detection performance by fixing the TR rank. Finally, we analyze the influence of the TR rank on the anomaly detection performance by fixing the TV and sparse regularization parameters λ and β.

First, the parameters λ, β determine the weights of TV regularization and sparse regularization on the background tensor $\Phi[\mathcal{G}]$ and the anomaly tensor $\mathcal{E}$, respectively. Different combinations of λ and β will obtain various detection performances. In this article, to find the optimal combinations of λ and β, these two parameters are both searched from $\{0.01, 0.05, 0.1, 0.5, 1, 5, 10\}$, where TR rank is fixed as $(r_1, r_2, r_3) = (4, 4, 4)$. The $\mathrm{AUC}_{(P_D,P_F)}$ values of the six datasets under different values of (λ, β) are displayed in Fig. 6. For the HYDICE dataset shown in Fig. 6(a), the robustness of the anomaly detection performance varies moderately with the parameters λ and β, and the $\mathrm{AUC}_{(P_D,P_F)}$ generally shows a decreasing trend with the increase of λ. The changing trend of $\mathrm{AUC}_{(P_D,P_F)}$ with the rise of β is not apparent. We get the best value of $\mathrm{AUC}_{(P_D,P_F)}$ when $\lambda = 0.1$ and $\beta = 10$. For the SpecTIR dataset shown in Fig. 6(b), the robustness of the anomaly detection performance to the parameters λ and β is excellent, and the $\mathrm{AUC}_{(P_D,P_F)}$ hardly changes with the changes of λ and β. We get the best value of $\mathrm{AUC}_{(P_D,P_F)}$ when $\lambda = 0.1$ and $\beta = 10$. For the urban-3 dataset shown in Fig. 6(c), the robustness of the anomaly detection performance to the parameter λ and β is also moderate. The value of $\mathrm{AUC}_{(P_D,P_F)}$ decreases with the increase of λ when $\beta = 0.01$,

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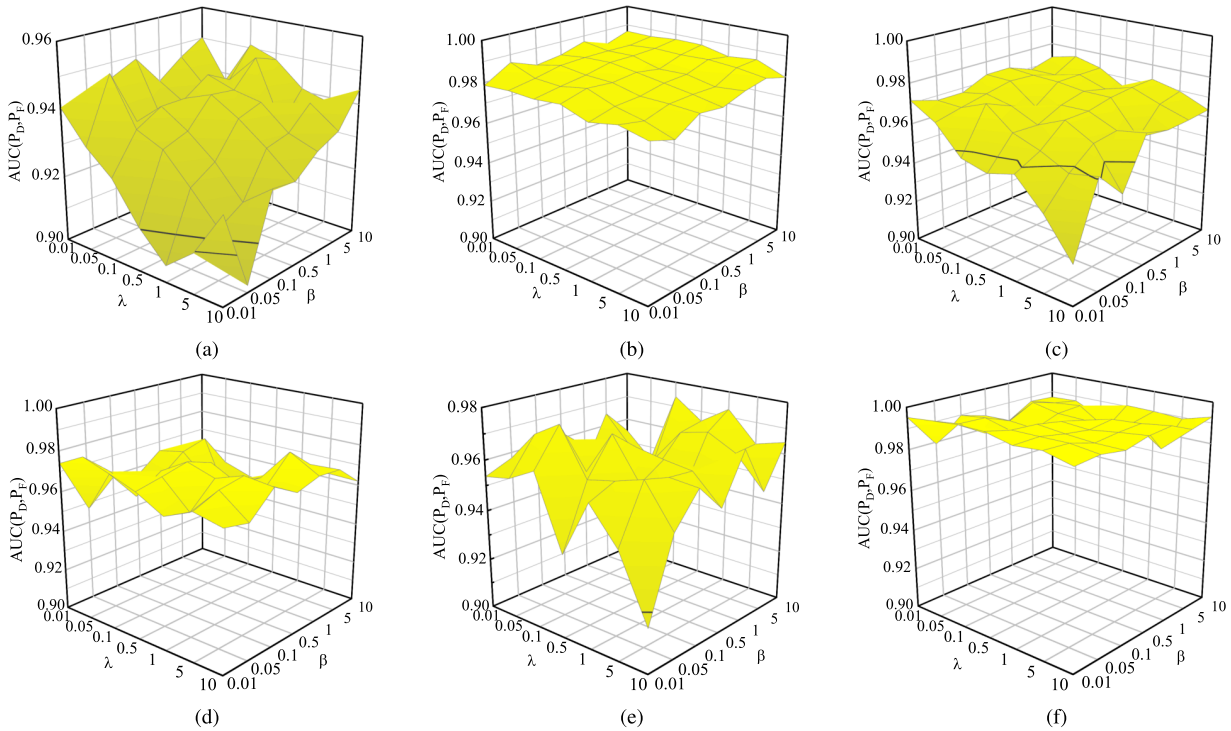


Fig. 6. Sensitivity analysis of parameters (λ, β). (a) HYDICE dataset. (b) SpecTIR dataset. (c) Urban-3 dataset. (d) Urban-5 dataset. (e) Airport-2 dataset. (f) Airport-4 dataset.

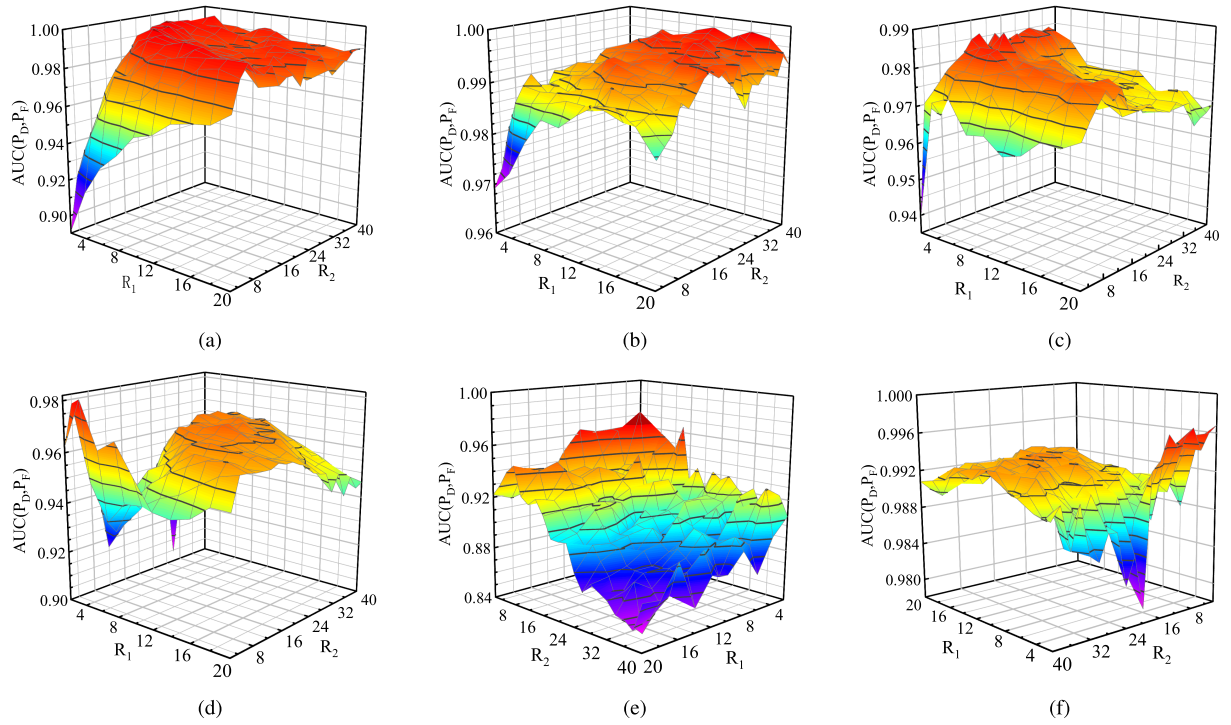


Fig. 7. Sensitivity analysis of the TR rank (r_1, r_2). (a) HYDICE dataset. (b) SpecTIR dataset. (c) Urban-3 dataset. (d) Urban-5 dataset. (e) Airport-2 dataset. (f) Airport-4 dataset.

the value of $\text{AUC}_{(P_D,P_F)}$ hardly changes with the changes of λ and β when $\beta > 0.01$, and we get the best value of $\text{AUC}_{(P_D,P_F)}$ when $\lambda = 0.05$ and $\beta = 10$. For the urban-5 dataset shown in Fig. 6(d), the robustness of the anomaly detection performance to the parameters λ and β is excellent. The $\text{AUC}_{(P_D,P_F)}$ almost hardly changes with the changes of λ and β. We get the best value of $\text{AUC}_{(P_D,P_F)}$ when $\lambda = 1$ and $\beta = 5$. For the airport-2 dataset shown in Fig. 6(e), the robustness of the anomaly detection performance to the parameter λ and β is also moderate. The value of $\text{AUC}_{(P_D,P_F)}$ decreases with the

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increase of λ, when $\beta = 0.01$. the value of $\text{AUC}_{(P_D,P_F)}$ hardly changes with the changes of λ and β, and we get the best values of $\text{AUC}_{(P_D,P_F)}$ when $\lambda = 0.1$ and $\beta = 0.05$. For the airport-4 dataset shown in Fig. 6(f), the anomaly detection performance is also robust to the combination of different values of λ and β, since the value of $\text{AUC}_{(P_D,P_F)}$ hardly changes with the change of λ and β. For this dataset, we get the best value of $\text{AUC}_{(P_D,P_F)}$ when $\lambda = 0.1$ and $\beta = 0.01$. Based on the above analysis, the six datasets can achieve good detection performance under different combinations of λ and β. However, individual tuning of λ and β is necessary to achieve the best detection performance for the different datasets.

Second, TR rank is also an essential factor that determines the detection performance of the proposed method. In this article, TR rank contains three parameters r_1, r_2, and r_3. The estimation of these parameters is still an open problem. If the value of the TR rank is too small, the low-rank property of the background in the three dimensions cannot be well represented. However, suppose the value of the TR rank is too large, the complexity of the calculation will be increased. Thus, choosing a suitable value of the TR rank is very important to determine the performance of the algorithm. From [53], we know that $r_1 \times r_2$, $r_2 \times r_3$, and $r_3 \times r_1$ are the bounded of $\text{rank}(\mathbf{B}_{\langle 1\rangle})$, $\text{rank}(\mathbf{B}_{\langle 2\rangle})$, and $\text{rank}(\mathbf{B}_{\langle 3\rangle})$, respectively. To simplify the complexity of the analysis of the parameters, we choose $r_1 = r_3$ to maintain the consistency of the spatial TR core tensors as shown in Fig. 1. Meanwhile, the background tensor $\mathcal{B}$ has a larger spatial rank and a smaller spectral rank. Thus, r_1 is range from 2 to 20 with two intervals, and r_2 is range from 2 to 40 with two intervals. The $\text{AUC}_{(P_D,P_F)}$ of the six datasets with different values of r_1 and r_2 are displayed in Fig. 7 with (λ, β) being $(0.1, 10)$, $(0.1, 10)$, $(0.05, 10)$, $(5, 1)$, $(0.01, 0.05)$, and $(0.1, 0.01)$ for the HYDICE, SpecTIR, urban-3, urban-5, airport-2, and airport-4, respectively. For the HYDICE, SpecTIR, urban-3, urban-5, and airport-4 datasets shown in Fig. 7, the detection performance is robust to the TR rank since the value of AUC does not change significantly with the change in TR rank. In contrast, the detection performance decrease when $r_1 > 10$ and $r_2 > 22$ for the airport-2 dataset. Specifically, for the HYDICE dataset shown in Fig. 7(a), we can get the best $\text{AUC}_{(P_D,P_F)}$ when $r_1 = 6$ and $r_2 = 16$. For the SpecTIR dataset, we can get the best $\text{AUC}_{(P_D,P_F)}$ when $r_1 = 16$ and $r_2 = 40$. For the urban-3 dataset, we can get the best $\text{AUC}_{(P_D,P_F)}$ when $r_1 = 6$ and $r_2 = 16$. For the urban-5 dataset, we can get the best $\text{AUC}_{(P_D,P_F)}$ when $r_1 = 4$ and $r_2 = 2$. For the airport-2 dataset, we can get the best $\text{AUC}_{(P_D,P_F)}$ when $r_1 = 4$ and $r_2 = 6$. For the airport-4 dataset, we can get the best $\text{AUC}_{(P_D,P_F)}$ when $r_1 = 2$ and $r_2 = 8$. From the simulation results above, it can be seen that for most datasets, the anomaly detection performance of the proposed algorithm is robust to the combination of r_1 and r_2. Of course, r_1 and r_2 need to be fine-tuned on different datasets to obtain the best performance.

V. Conclusion

In this article, we propose a novel TRDFTVAD. In the proposed method, the low-rank background tensor is decomposed into several factor tensors to exploit the low-rank property of the background existing in the spatial and spectral dimensions. Then, the TV regularization is designed on each factor tensor to exploit the piecewise smoothness of the background existing both in the spatial and spectral dimensions. Next, the $l_{2,1}$ regularization is designed on the anomaly tensor to exploit the group sparsity. Finally, an ADMM approach is introduced to solve the proposed model. The proposed method fully uses the low-rank and piecewise smoothness properties of the background in both spatial and spectral dimensions and thus can improve anomaly detection performance. Experiments on several real datasets demonstrate the effectiveness of the proposed method. In future work, we will focus on reducing the complexity of the proposed method and improving the robustness of the proposed method to parameter initialization.

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